Executive Summary:

Due to the increased maintenance expenditures and potential dangers posed by unattended trees, urban tree management is becoming an important responsibility for developed city governments. The public sector/smart city domain specifically categorizes this project, with a focus on urban environmental management. Using tree data supplied by Dublin City Council, this project uses machine learning techniques to forecast the state of the public trees in Dublin.

Predicting the state of a tree from data collected about its surroundings, including its age, species, size, and closeness, is the crux of this multi-class classification challenge. A basic model called 'Logistic Regression' and an advanced ensemble model called 'Random Forest' were both created and tested.

An appropriate metric for unbalanced data is by using the F1-macro score, which is used to assess the results and demonstrate that the Random Forest model surpasses logistic regression. The results show how smart city environments can benefit from predictive analytics in terms of proactive, data-driven decision-making, operational cost reduction, and public safety.

Introduction:

Problem Statement:

Dublin City Council is obligated to protect the numerous public trees in the city by physically inspecting them regularly. Currently, the inspections rely heavily on reactive reporting and manual scheduling. The cumbersome procedure might potentially waste time and energy because branches can break off, or the tree can fall, especially since certain trees age more quickly than others due to structural and environmental factors. However, this project aims to apply a predictive system that can identify trees at risk, i.e., more likely to be in danger, so the council can schedule inspections and maintenance, thereby reducing extreme overhead expenditures, optimizing inspection schedules, and improving public safety. The problem is modeled with a multi-class classification task which is important for determining the health status of the trees based on the observed available attributes.

Data Source

The dataset used in this project is the Dublin City Council Tree Dataset, sourced from the open data portal of Ireland. Each record represents an individual tree managed by the council.

Key attributes include: - Species - Age group - Height range - Spread range - Stem diameter range - Proximity to infrastructure - Area/location - Tree condition (target variable)

The dataset is tabular and contains both categorical and numeric-like range features.

https://data.smartdublin.ie/dataset/tree-dcc

3. Data Analysis.

This project seeks to utilize exploratory data analysis, revealing the categorical and numerical variables in the data. It is noteworthy that the missing values be cleaned to ensure accuracy and consistency in the analysis. Additionally, the target variable {Condition} is imbalanced, frequently showing some health conditions more than others.

Visualizations:

Exploratory data analysis resulted in three important visualizations. There was an obvious class imbalance in the distribution of tree conditions, with certain condition types appearing considerably more often than others. Dataset dominance by a small number of species, as revealed by an analysis of the most prevalent tree species, suggests little species variety across numerous places. Tree maturity levels, exposure to nearby infrastructure, and environmental variables were all illuminated by the distributions of age groups and proximity categories. It was made sure that the modeling technique took data imbalance and structural aspects of the dataset into proper consideration by combining these visual findings, which inspired the preprocessing procedure and affected the selection of evaluation metrics.

4. Methodology

Preprocessing Steps

The dataset underwent a standardized preprocessing pipeline to guarantee trustworthy, impartial, and robust model training. Each feature type had its own strategy for dealing with missing data. To maintain consistent patterns in the data, we imputed categorical variables using the value that appeared most often. The median was used to impute numerical features since it is stable in the face of outliers and preserves the data's overall distribution.

We next divided the dataset in half, setting aside 20% to be used for evaluation and 80% for training the model. The original class distribution of the target variable was preserved by the use of stratified sampling, which reduced the danger of biased model performance due to class imbalance.

In order to make a number of important variables more user-friendly, feature engineering was performed. Originally, textual ranges were used to encode attributes relating to tree size, such as height, spread, and stem diameter. The models were able to successfully learn from continuous size-related data when they were converted to numerical midpoint values. Also, to make training the model more stable and less noisy, we combined all the rare tree condition categories into one "Other" class.

5. Model Implementation

Model 1: Logistic Regression

The simplicity, computational efficiency, and ease of comprehension of Logistic Regression led to its selection as the baseline model. As a result, it's easy to compare more complicated models to this one. The model was set up to guarantee convergence with a maximum of 1000 training iterations. Feature scaling for numerical variables and one-hot encoding for categorical features were part of the preprocessing pipeline that was used for training. This model set the bar for more advanced methods to be measured against.

Model 2: Random Forest Classifier

We opted for the Random Forest classifier since it is a sophisticated model that can capture complicated interactions between features and non-linear correlations. To ensure repeatability, the model was set up using 200 decision trees. To tackle the issue of class imbalance in the target variable, balanced class weighting was implemented. The Random Forest model and the Logistic Regression model used the same preprocessing pipeline during training to ensure consistency and fair comparison.

6. Discussion and Results

Comparative Results:

We used the F1-macro score and accuracy as our metrics to measure the model's performance. The F1-macro score is well-suited for imbalanced datasets because it gives equal weight to each class, whereas accuracy measures the total correctness of predictions. In comparison to the Logistic Regression model, the Random Forest model outperformed it in every category of tree condition, including minority classes, as evidenced by its better F1-macro score.

1. Comparing Models

The computational efficiency and interpretability of Logistic Regression were high, but it was not able to capture complicated, non-linear correlations in the data. When it came to under-represented tree condition classes, however, the Random Forest model showed better predictive potential. For practical deployment in the real world, its capacity to represent non-linear patterns and interactions is a major plus.

2. Business Impact Analysis

The Random Forest model has the potential to greatly improve Dublin City Council's operations. The council can save money on emergency maintenance, make inspection scheduling more efficient, and make the public safer by proactively identifying trees that are likely to be in bad health. This proves that machine learning is useful for smart city asset management decision-support.

7. Conclusions and Future Work

Conclusions

The successful utilization of machine learning approaches to forecast tree health issues from real-world urban data is showcased in this study. Random Forest was determined to be the best model after extensive preprocessing, focused feature engineering, and examination of competing models.

A data-driven, scalable answer to the problems of urban tree management is offered by the suggested method.

Future Work

To further understand the environmental factors that affect tree health, it would be beneficial to incorporate geographical and weather-related data in future updates. Further improvement of predictive accuracy may be possible with advanced modeling approaches like Gradient Boosting. Furthermore, high-risk misclassifications might be prioritized using cost-sensitive learning techniques, and the model could be implemented in a real-time decision support system for practical usage.